MAT 1272

TI-84 Quick Reference for the Final Exam

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TI-84 Commands You Should Know

For the final exam, know how to use:

1. normalcdf
2. invNorm
3. binompdf
4. binomcdf
5. invT
6. Z-Test
7. T-Test
8. Chi-Square GOF-Test
9. Chi-Square Test

To find a probability:

$$P(a < X < b)$$

use:

```
normalcdf(lower, upper, mean, standard deviation)
```

Example:

```
normalcdf(60,70,64,3)
```

To find a value corresponding to a percentile:

```
invNorm(area, mean, standard deviation)
```

Example:

```
invNorm(0.95,64,3)
```

Binomial Distribution

Exactly x successes:

`binompdf(n,p,x)`

Example:

`binompdf(10,0.30,4)`

At most x successes:

`binomcdf(n,p,x)`

Example:

`binomcdf(10,0.30,4)`

More than x successes:

$1 -$ `binomcdf(n,p,x)`

Critical z-Values

Use:

```
invNorm(area)
```

Right-tailed test:

$$\alpha = 0.05$$

```
invNorm(0.95)
```

$$z_c = 1.645$$

Two-tailed test:

$$\alpha = 0.05$$

```
invNorm(0.975)
```

$$z_c = 1.96$$

Critical t-Values

Use:

```
2nd -> VARS -> invT
```

Example:

$$\alpha = 0.05, \quad df = 15$$

Right-tailed:

```
invT(0.95,15)
```

$$t_c = 1.753$$

Two-tailed:

```
invT(0.975,15)
```

$$t_c = 2.131$$

Hypothesis Testing

Population mean, σ known:

```
STAT -> TESTS -> Z-Test
```

Population mean, σ unknown:

```
STAT -> TESTS -> T-Test
```

The calculator reports:

- Test statistic
- p-value

Chi-Square Goodness-of-Fit Test

1. Enter observed counts in L_1.
2. Enter expected counts in L_2.
3. Press STAT.
4. Select TESTS.
5. Choose Chi-Square GOF-Test.
6. Enter degrees of freedom.
7. Calculate.

Record:

- p-value
- degrees of freedom
- conclusion

Chi-Square Test of Independence

1. Press 2nd MATRIX.
2. EDIT Matrix A.
3. Enter the contingency table.
4. Press STAT.
5. Select TESTS.
6. Choose Chi-Square Test.
7. Calculate.

Record:

- p-value
- degrees of freedom
- conclusion

Decision Rule

For every hypothesis test:

$$\text{If } p < \alpha,$$

Reject H_0

$$\text{If } p \geq \alpha,$$

Fail to Reject H_0

Always write a conclusion in context.

Good Luck on the Final!

Know the calculator commands.

Know how to interpret p-values.

Know how to write conclusions.